SOLUTIONS: STABILITY AND CONTROL OF NONLINEAR SYSTEMS MASTER IN CONTROL

1. Exercise

a) The function $f(x)$ that defines the system's equations is polynomial in both variables. Hence it is locally Lipschitz and by Caratheodory's theorem solutions exist and are unique (over their maximal time-interval of definition). [2]

b) The nullclines of the vector-field can be computed as shown below:

$$\mathscr{N}_1 = \{x : x_2 = x_1^3 - 2x_1\}.$$

This is the equation of a cubic which intersects the x_1 axis in $\pm\sqrt{2}$ and 0. Moreover, the slope in 0 of the nullcline is -2. All solutions above the cubic move eastwards and those below west. For the second nullcline we see:

$$\mathscr{N}_2 = \{x : x_1 = x_2^3 - 2x_2\}.$$

The second nullcline is therefore of the same shape as nullcline $\mathscr{N}_1$, but symmetric with respect to the diagonal. All solutions to the right of the cubic move south and those to the left move north. [3]

c) There are 8 regions delimited by the nullclines. The direction of the vector-field in each one of them is shown in Fig. 1.1. [3]

d) From the direction of the arrows we notice that all regions whose interior intersects the diagonal (dotted line) are forward invariant (while the rest are not). [2]

e) To find equilibria of the system we look at nullclines intersections. There are 5 intersections. Due to symmetry of the nullclines these are either along the diagonal, or on the anti-diagonal. Hence we can find them using the extra constraint $x_1 = \pm x_2$. On the diagonal we see that $x_1 = x_1^3 - 2x_1$ yields $x_1 = 0$ and $x_1 = \pm\sqrt{3}$. Thus three equilibria are in $[0,0]', [\sqrt{3},\sqrt{3}]', [-\sqrt{3},-\sqrt{3}]'$. Instead for $x_1 = -x_2$ we see that $-x_1 = x_1^3 - 2x_1$ yields $x_1 = \pm 1$, thus giving 2 more equilibria in $[1,-1]$, $[-1,1]$. [2]

f) To linearize the system we first compute the Jacobian matrix $\frac{\partial f}{\partial x}$ at a generic value of x.

$$\frac{\partial f}{\partial x}(x) = \begin{bmatrix} -3x_1^2+2 & 1 \\ 1 & -3x_2^2+2 \end{bmatrix}.$$

At the equilibrium $x_{e1} = [0,0]'$ we see that:

$$\frac{\partial f}{\partial x}(x_{e1}) = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Its characteristic polynomial is $\chi_1(s) = s^2 - 4s + 3$, whose roots are 1 and 3. Two positive eigenvalues correspond to the phase potrait of an unstable node (hyperbolic). At the equilibria $x_{e2} = [\sqrt{3},\sqrt{3}]'$ and $x_{e3} = -[\sqrt{3},\sqrt{3}]'$ we have the Jacobian:

$$\frac{\partial f}{\partial x}(x_{e2}) = \begin{bmatrix} -7 & 1 \\ 1 & -7 \end{bmatrix}.$$

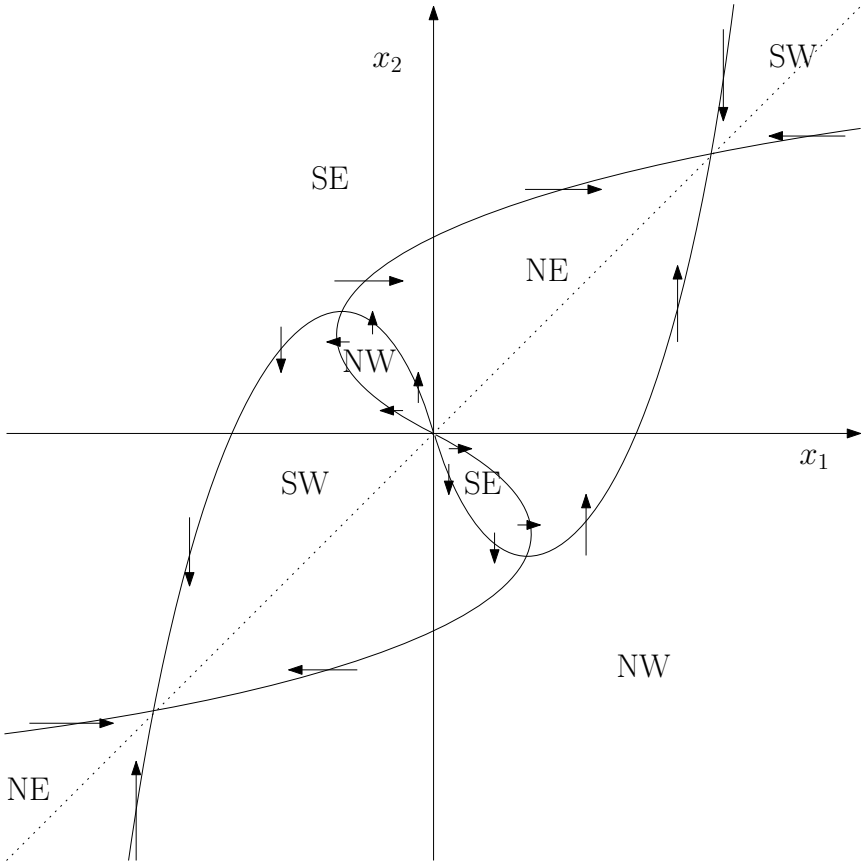


Figure 1.1 Nullclines and vector-field orientations

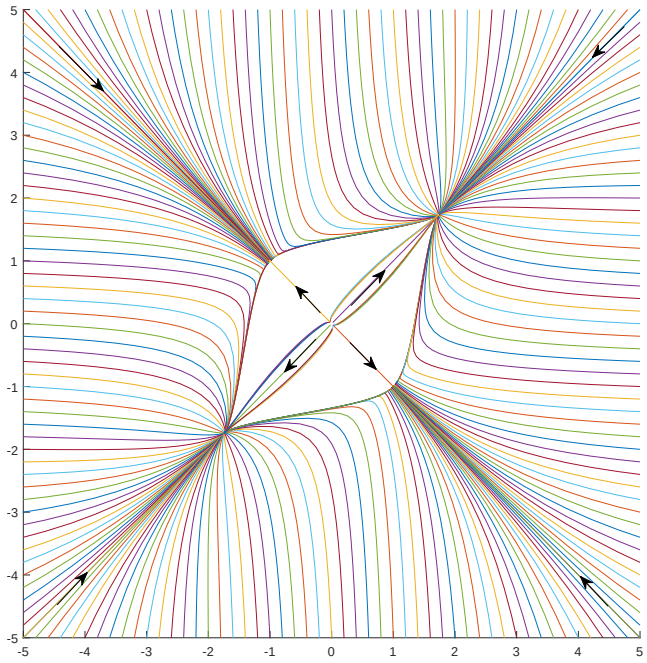


Figure 1.2 Phase portrait

Its characteristic polynomial is $\chi_2(s) = \chi_3(s) = s^2 + 14s + 48$ with roots in -8 and -6. Two negative roots correspond to the phase portrait of a stable node (hyperbolic). At the equilibria $x_{e4} = [1,-1]'$ and $x_{e5} = [-1,1]'$ the Jacobian takes the following form:

$$\frac{\partial f}{\partial x}(x_{e4}) = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Its characteristic polynomial is $\chi_4(s) = \chi_5(s) = s^2 + 2s$ with eigenvalues in 0, -2. This is not hyperbolic. From the nullclines we see that the local phase portrait appears similar to that of a saddle point. [3]

g) To show that the diagonal is invariant it is enough to see that for $x_1 = x_2$, it holds $\dot{x}_1 = \dot{x}_2$. This is trivially seen given the symmetry of the equations. [3]

h) The global phase portrait is sketched in Fig. 1.2. [2]

2. Exercise

a) We may take P to be the identity. Indeed:

$$A_1' + A_1 = \begin{bmatrix} -4 & 1 \\ 1 & -2 \end{bmatrix} < 0$$

and similarly:

$$A_2' + A_2 = \begin{bmatrix} -2 & 1 \\ 1 & -4 \end{bmatrix} < 0.$$

Two marks allocated for writing up systems of conditions on P and three marks for finding solution. [5]

b) To see that $x = 0$ is an equilibrium for all $d \in \mathbb{R}$ it is enough to compute:

$$f(0,d) = dA_1 \cdot 0 + (1-d)A_2 \cdot 0 = 0.$$

[1]

c) Next, we choose as a Lyapunov function $V(x) = x'x$. Its derivative along solutions of the system is:

$$\begin{aligned} \dot{V} = \dot{x}'x + x'\dot{x} &= dx'(A_1' + A_1)x + (1-d)x'(A_2' + A_2)x \\ &\leq d\bar{\lambda}(A_1' + A_1)x'x + (1-d)\bar{\lambda}(A_2' + A_2)x'x = (-3+\sqrt{2})x'x. \end{aligned}$$

where $\bar{\lambda}(M)$ denotes the largest (real) eigenvalue of M. In particular, $\bar{\lambda}(A_2' + A_2) = \bar{\lambda}(A_1' + A_1) = -3 + \sqrt{2}$. [2]

Hence $V(x)$, positive definite and radially unbounded, has a negative definite derivative (uniformly for $d \in [0,1]$) and this shows that the origin is a UGAS equilibrium. [2]

d) To study the scalar system $\dot{z} = -4z^3 + z$ we first compute its equilibria:

$$-4z^3 + z = 0 \Rightarrow z \in \{\pm 1/2, 0\}.$$

Moreover, $-4z^3 + z < 0$ for $z > 1/2$, and $z \in (-1/2, 0)$, while $-4z^3 + z > 0$ for $z < -1/2$ and $z \in (0, 1/2)$. In particular then, exploiting monotonicity of solutions for scalar systems, we see that $\omega(z_0) = 0$ for $z_0 = 0$, $\omega(z_0) = \{1/2\}$ for $z_0 > 0$ and $\omega(z_0) = \{-1/2\}$ for $z_0 < 0$. [3]

e) The equations of the cascaded system read:

$$\begin{aligned} \dot{z} &= -4z^3 + z \\ \dot{x} &= (1+z)A_1 x/2 + (1-(1+z)/2)A_2 x. \end{aligned}$$

[2]

f) Notice that, when the state of system Σ_2 is at equilibria $-1/2, 0, 1/2$, the corresponding output values are $1/4, 2/4, 3/4$ (respectively). Therfore, the previous analysis implies that regardless of the initial condition z_0, there exists a finite time T such that $y(t) \in [0,1]$ for all $t \geq T$. (Notice that solutions of the cascaded system are defined for all times since f is linear in x). Hence, regardless of z_0 and x_0 there exists a finite time T such that $d(t) \in [0,1]$ for all $t \geq T$. Because of UGAS, there exists β of class $\mathscr{KL}$ such that:

$$|x(t)| \leq \beta(|x(T)|, t-T) \qquad \forall t \geq T.$$

As a consequence, $x(t) \to 0$ as $t \to +\infty$. Overall the ω-limit set of the cascaded system is still a singleton for each initial condition. In particular, equal to $\{[-1/2, 0, 0]'\}, \{[0,0,0]\}$ and $\{[1/2,0,0]\}$ depending on the initial value for z_0, respectively negative, 0 or positive. Three marks allocated for correct answer and two for rigorous justification. [5]

3. Exercise

a) For a SISO control affine system,

$$\dot{x} = f(x) + g(x)u, \quad y = h(x),$$

passivity and losslessness amount to the following two conditions:

$$\frac{\partial S}{\partial x}(x) \cdot f(x) = 0$$

$$\frac{\partial S}{\partial x}(x) \cdot g(x) = h(x).$$

Hence we may require that:

$$\frac{\partial S}{\partial x_1}(x) 4x_2 - \frac{\partial S}{\partial x_2}(2x_1 + 2\sin(x_1)\cos(x_1)) = 0$$

$$\frac{\partial S}{\partial x_2}(x) = y = 4x_2.$$

[3]

Integrating the second one with respect to x_2 we see that $S(x)$ needs to be of the following form:

$$S(x) = 2x_2^2 + \tilde{S}(x_1),$$

for some $\tilde{S}$ function of x_1 only. Similarly, substituting $4x_2$ for $\frac{\partial S}{\partial x_2}$ in the first equation we see that we can take:

$$\frac{\partial S}{\partial x_1}(x) = 2x_1 + 2\sin(x_1)\cos(x_1).$$

Integration of the latter yields:

$$S(x) = x_1^2 + \sin^2(x_1) + \hat{S}(x_2),$$

for some suitable function $\hat{S}$ of x_2. Combining the previous expressions we see that:

$$S(x) = x_1^2 + \sin^2(x_1) + 2x_2^2$$

is a suitable storage function. [2]

b) We propose the following static output feedback:

$$u = -\frac{e^y - 1}{e^y + 1} := k(y).$$

Notice that this is an increasing function of y, such that $|k(y)| \leq 1$ for all $y \in \mathbb{R}$. Moreover $k(y) > 0$ iff $y < 0$ and $k(y) < 0$ iff $y > 0$. [2]

Overall, we let $V(x) := S(x)$ be a candidate Lyapunov function. We see that V is positive definite and radially unbounded, since $V(x) \geq x_1^2 + 2x_2^2$. For its derivative we have:

$$\dot{V} = [2x_1 + 2\sin(x_1)\cos(x_1), 4x_2][f(x) + g(x)u] = yu = yk(y) \leq 0.$$

Hence, $\dot{V}$ is negative semidefinite and $\text{Ker}[\dot{V}] = \{x : x_2 = 0\}$. Any invariant set contained in $\text{Ker}[\dot{V}] = \{x : x_2 = 0\}$ is therefore also contained in

$$\{x : x_2 = 0 \text{ and } -2x_1 - 2\sin(x_1)\cos(x_1) = 0\} = \{0\}.$$

Hence, the system is GAS by Lasalle's invariance principle. [3]

c) Considering the equation for $\dot{S}$ we see that:

$$\dot{S} = yu.$$

We propose the feedback:

$$u = y(S_{ref} - S(x)).$$

[2]

Consider next the Lyapunov function $W(x) := (S(x) - S_{ref})^2$. We see that:

$$\dot{W} = 2(S(x) - S_{ref})yu = -2y^2(S(x) - S_{ref})^2 \leq 0.$$

[3]

d) Since $\dot{W} \leq 0$ we have that $W(x_{cl}(t))$ is non-increasing along systems solutions. Moreover it is bounded below. Hence its limit as $t \to +\infty$ exists. [2]
Notice that the ω-limit set of solutions is non-empty and invariant and therefore is contained in a level set of W. This implies that solutions in $\omega(x_0)$ have zero $\dot{W}$. This implies zero y or zero $S(x) - S_{ref}$. The first case can be ruled out since y identically zero implies u identically zero and x_1 identically zero, which only occurs when $x_1(0) = x_2(0) = 0$ and $S_{ref} = 0$. [3]

4. Exercise

a) To compute the relative degree we consider the output and its Lie derivatives until the input u appears in the corresponding expressions:

$$\begin{aligned} y &= x_1 + x_2 \\ \dot{y} &= \dot{x}_1 + \dot{x}_2 = x_1 - x_2 \\ \ddot{y} &= \dot{x}_1 - \dot{x}_2 = 2\text{atan}(x_1) + 2(2+\cos(x_2))u. \end{aligned}$$

[2]

Since the input u only appears after 2 derivations, the relative degree is 2. Moreover, its coefficient $2(2+\cos(x_2))$ is non zero for all $x \in \mathbb{R}^3$ and therefore the relative degree is globally defined. [2]

b) We choose the feedback law:

$$u = \frac{v - 2\text{atan}(x_1)}{2(2+\cos(x_2))}.$$

With this choice the input-output equations read: $\ddot{y} = v$, which is indeed the equation of a double integrator. [3]

c) We use the following variables as a set of independent coordinates:

$$\xi_1 = y, \qquad \xi_2 = \dot{y}, \qquad z = x_3.$$

Accordingly we have the equations:

$$\begin{aligned} \dot{\xi}_1 &= \xi_2 \\ \dot{\xi}_2 &= v \\ \dot{z} &= -z^3 + \xi_1 \\ \\ y &= \xi_1 \end{aligned}$$

[3]

The above equations are in normal form. Moreover the equation for $\dot{z}$ describes the so called internal dynamics. [2]

d) The internal dynamics are one-dimensional, and fulfill the equation:

$$\dot{z} = -z^3 + \xi_1$$

where ξ_1 is regarded as the exogenous input signal. These equations define an ISS system. [2]
Indeed we can take the ISS Lyapunov function: $V(z) = z^2/2$ which is positive definite and radially unbounded. Along solutions of the system we see that:

$$\dot{V} = z(-z^3 + \xi_1) = -z^4 + z\xi_1 \leq -|z|^4 + |z||\xi_1|.$$

Assume $|z| \geq \chi(|\xi_1|) := \sqrt[3]{2|\xi_1|}$. Notice that χ is a class $\mathscr{K}_\infty$ function. We see that:

$$|z| \geq \chi(|\xi_1|) \Rightarrow |z|^3 \geq 2|\xi_1| \Rightarrow |z|^4/2 \geq |z||\xi_1| \Rightarrow \dot{V} \leq -|z|^4/2.$$

Hence V is an ISS Lyapunov function of type 2, and ISS of the internal dynamics hold. [2]

e) We may pick the stabilizing feedback as $v = -2\dot{y} - y$. [2]
This induces a cascade of a linear systems with two poles in -1 which is forcing the internal dynamics. Since the internal dynamics are ISS the cascaded interconnection is also Globally Asymptotically Stable. [2]